

Control Variates: Foundations and Intuition

Monte Carlo Statistical Methods

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1 Universal Linear Decomposition

Consider two dependent random variables X and Y . We know only their basic second-order information:

$$E[X], \quad E[Y], \quad \text{Var}(X), \quad \text{Var}(Y), \quad \text{Cov}(X, Y).$$

We do not know the joint distribution $f_{X,Y}(x, y)$ or the true form of the relationship between X and Y . In general, Y could be linear,

$$Y = aX + b,$$

nonlinear,

$$Y = f(X),$$

or noisy,

$$Y = f(X) + \eta.$$

All we can assume is that X and Y are statistically dependent.

1.1 Goal

With so little structural information, the question is:

How can we describe the relationship between X and Y using only what we know?

We cannot identify f , the noise η , or any nonlinear structure. However, there is one relationship that always exists.

1.2 Taylor View

Taylor's idea is simple: if a function is unknown, begin with its best linear approximation:

$$f(x) \approx f(x_0) + f'(x_0)(x - x_0).$$

The linear term captures the dominant trend, and everything else becomes a remainder.

We apply the same idea to random variables. Even without knowing the true mechanism linking X and Y , we can approximate Y by the best possible linear function of X and treat everything else as a residual. This residual collects all nonlinear behavior and all noise in Y that cannot be explained by X , and it will turn out to be uncorrelated with X by construction.

This leads naturally to the universal linear decomposition that follows.

2 Best Linear Approximation of Y by X

Given dependent random variables X and Y , we know

$$E[X], \quad E[Y], \quad \text{Var}(X), \quad \text{Var}(Y), \quad \text{Cov}(X, Y).$$

We approximate Y by a linear function of X and with all remaining nonlinearity and noise collected into a residual ϵ :

$$Y = aX + b + \epsilon.$$

2.1 Deriving a and b

We choose a and b by minimizing

$$\min_{a,b} E[(Y - (aX + b))^2].$$

Optimal b Set the derivative w.r.t. b to zero:

$$E[Y - aX - b] = 0 \quad \Rightarrow \quad b = E[Y] - a E[X].$$

Optimal a Set the derivative w.r.t. a to zero:

$$E[X(Y - aX - b)] = 0.$$

Substitute the optimal b :

$$E[(X - E[X])(Y - E[Y])] - a E[(X - E[X])^2] = 0,$$

so

$$a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}.$$

2.2 Residual Properties

Define

$$\epsilon = Y - (aX + b).$$

Then:

Zero Mean

$$E[\epsilon] = 0.$$

Orthogonality

$$E[X\epsilon] = 0 \quad \Rightarrow \quad \text{Cov}(X, \epsilon) = 0.$$

2.3 Summary of Findings

- The best linear approximation of Y using X is

$$Y = aX + b + \epsilon.$$

- The coefficients depend only on second-order statistics:

$$a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}, \quad b = E[Y] - a E[X].$$

- The residual ϵ captures all variation in Y not explained linearly by X :

$$\begin{aligned} E[\epsilon] &= 0, \\ \text{Cov}(X, \epsilon) &= 0. \end{aligned}$$

Thus $aX + b$ is the best linear part of Y explained by X , and ϵ contains everything else.

3 Variance Decomposition

Starting from the decomposition

$$Y = aX + b + \epsilon,$$

subtracting expectations gives

$$Y - E[Y] = a(X - E[X]) + \epsilon.$$

Taking variances yields the key identity

$$\boxed{\text{Var}(Y) = a^2 \text{Var}(X) + \text{Var}(\epsilon)}$$

This shows that the variability of Y arises from two uncorrelated components:

$$\underbrace{a^2 \text{Var}(X)}_{\text{variance in } Y \text{ explained by } X} + \underbrace{\text{Var}(\epsilon)}_{\text{remaining variance not explained by } X}.$$

3.1 Connection with Correlation

Since

$$a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

we obtain

$$a^2 \text{Var}(X) = \frac{\text{Cov}(X, Y)^2}{\text{Var}(X)} = \rho^2 \text{Var}(Y),$$

where

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}$$

is the correlation coefficient between X and Y .

Therefore

$$\boxed{\text{Var}(Y) = \rho^2 \text{Var}(Y) + (1 - \rho^2) \text{Var}(Y)}.$$

3.2 Meaning of ρ^2

The quantity ρ^2 has a precise interpretation:

$$\boxed{\rho^2 = \frac{\text{variance of } Y \text{ that is explained by } X}{\text{total variance of } Y}}$$

Thus:

- ρ^2 is the percent of the variance of Y that is explained by changes in X .
- $1 - \rho^2$ is the percent of the variance of Y that arises from the residual noise ϵ .

3.3 Examples and Interpretation

We illustrate the meaning of ρ^2 through three fundamental cases. In each case, recall the variance split:

$$\text{Var}(Y) = \rho^2 \text{Var}(Y) + (1 - \rho^2) \text{Var}(Y).$$

1. **Case 1: $\rho^2 = 1$ (perfect linear explainability)**

$$\rho^2 = 1 \quad \Rightarrow \quad X \text{ explains 100\% of the variance of } Y.$$

Then

$$\text{Var}(\epsilon) = 0,$$

so the decomposition becomes

$$Y = aX + b.$$

There is no residual noise and no nonlinear component. The true relationship between X and Y is exactly linear. All variability in Y is inherited directly from X .

2. **Case 2: $\rho^2 = 0$ (no linear explainability)**

$$\rho^2 = 0 \quad \Rightarrow \quad X \text{ explains 0\% of the variance of } Y.$$

Here

$$a = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = 0,$$

which gives the best linear decomposition

$$Y = b + \epsilon.$$

This does *not* imply independence between X and Y . It only states that their dependence has no linear component. A nonlinear relationship may still exist. For example, if $Y = X^2$ with X symmetric about zero, then $\text{Cov}(X, Y) = 0$ even though Y is completely determined by X . In the linear decomposition $Y = b + \epsilon$, the entire nonlinear term X^2 is pushed into ϵ , so all of the variance of Y comes from the residual.

3. **Case 3: $\rho^2 = 0.85$ (partial explainability)**

$$\rho^2 = 0.85 \quad \Rightarrow \quad X \text{ explains 85\% of the variance of } Y.$$

The component

$$a(X - E[X])$$

accounts for most of the variability in Y , leaving only

$$(1 - 0.85) \text{Var}(Y) = 0.15 \text{Var}(Y)$$

for the residual noise ϵ .

The linear trend is dominant; the unexplained part is small.

4 The Control Variate Insight

From the universal decomposition

$$Y = aX + b + \epsilon, \quad b = E[Y] - aE[X],$$

we have the centered identity

$$Y - E[Y] = a(X - E[X]) + \epsilon.$$

This expresses every deviation of Y from its mean as a sum of two uncorrelated components:

- a predictable component driven by how much X differs from its mean,
- an uncorrelated residual ϵ .

The variance split

$$\text{Var}(Y) = a^2 \text{Var}(X) + \text{Var}(\epsilon)$$

shows that part of the variability in Y is inherited from the randomness in X , while the remainder comes from intrinsic noise in ϵ .

4.1 The Estimation Problem

In this foundational discussion we assume that $E[X]$ and a are known. This allows us to focus purely on the structural ideas behind variance reduction. In practice, a is estimated from the same samples, which is the topic of the next chapter.

We observe i.i.d. pairs (X_i, Y_i) and our goal is to estimate the expectation

$$E[Y].$$

The simple Monte Carlo estimator

$$\hat{\mu}_Y = \frac{1}{n} \sum_{i=1}^n Y_i$$

is unbiased, but its variance is

$$\text{Var}(\hat{\mu}_Y) = \frac{1}{n} \text{Var}(Y),$$

which can be large when $\text{Var}(Y)$ is large.

The decomposition above suggests a way to construct a new random variable with the same expectation as Y but with smaller variance.

4.2 Understanding a Single Observation

For any observed pair (X_i, Y_i) ,

$$Y_i - E[Y] = a(X_i - E[X]) + \epsilon_i.$$

Interpretation:

- $Y_i - E[Y]$ is the total deviation of the sample Y_i from its mean.

- $X_i - E[X]$ is the deviation of X_i from its mean.
- $a(X_i - E[X])$ is the portion of $Y_i - E[Y]$ caused solely by the deviation of X_i . This term is predictable because we observe X_i and we assume a and $E[X]$ are known.
- ϵ_i is the remaining intrinsic noise that cannot be explained by X .

Thus each deviation in Y_i can be separated into a predictable part and a residual unpredictable part.

4.3 Removing the Predictable Component

Subtract the X -driven deviation:

$$(Y_i - E[Y]) - a(X_i - E[X]) = \epsilon_i.$$

This suggests defining a new random variable

$$Z = Y - a(X - E[X]).$$

Then

$$E[Z] = E[Y], \quad \text{Var}(Z) = \text{Var}(\epsilon).$$

Using

$$\text{Var}(\epsilon) = (1 - \rho^2)\text{Var}(Y),$$

we see that Z has much smaller variance than Y whenever ρ^2 is large.

4.4 A Lower Variance Estimator

Given samples (X_i, Y_i) , define transformed samples

$$Z_i = Y_i - a(X_i - E[X]).$$

Each Z_i has expectation $E[Y]$ and variance $\text{Var}(\epsilon)$. Thus the natural estimator for $E[Y]$ is

$$\hat{\mu}_Z = \frac{1}{n} \sum_{i=1}^n Z_i.$$

Its variance is

$$\text{Var}(\hat{\mu}_Z) = \frac{1}{n} \text{Var}(\epsilon) = \frac{1}{n} (1 - \rho^2) \text{Var}(Y),$$

which is strictly smaller than $\text{Var}(\hat{\mu}_Y)$ whenever $\rho^2 > 0$.

In words: if part of the variability in Y is explained by X , and we observe X , we can subtract the predictable part of Y . What remains has much less randomness and therefore yields a more stable estimator of $E[Y]$.

4.5 Bridge to the Full Control Variate Method

In this foundation chapter we assumed that a and $E[X]$ are known. In practice a must be estimated from data and the subtraction coefficient is chosen optimally by minimizing variance. This leads to the formal control variate estimator, developed in the next chapter.